

Federated Continual Learning

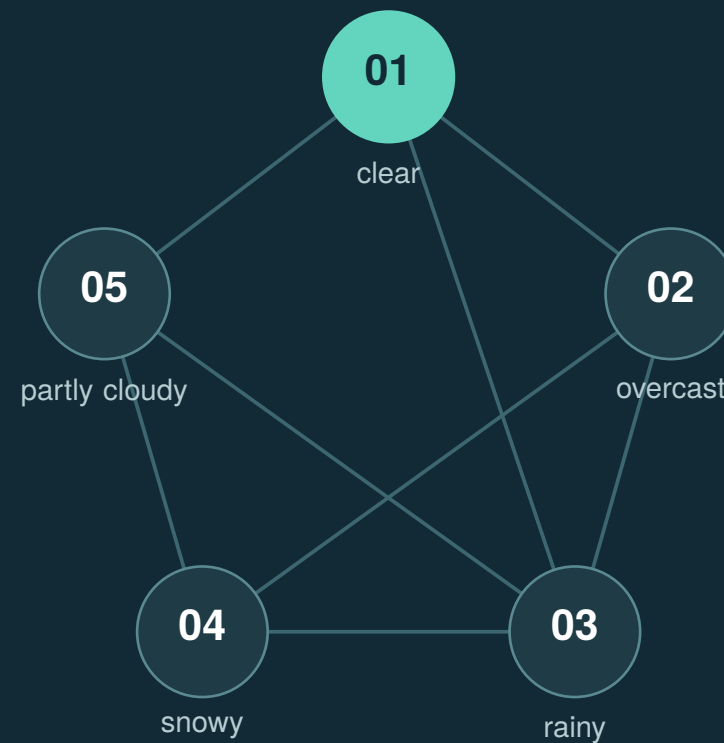
for Multi-Label Object Detection

BDD100K · Faster R-CNN
Weather-partitioned clients

01 PARTITION

02 TRAINING

03 EVALUATION



5 CLIENTS / PEER-TO-PEER

Three metadata fields. Two partition axes.

One image record exposes weather, scene and time of day; only two fields define this experiment.

RAW PER-IMAGE METADATA

```
"name": "0000f77c-6257be58.jpg",  
"attributes": {  
  "weather": "clear",  
  "scene": "city street",  
  "timeofday": "daytime"  
},
```

bdd100k_labels_images_train.json

CLIENT AXIS

weather

Five weather-specific clients.

TASK AXIS

timeofday

Three sequential tasks per client.

NOT USED

scene

Present in the metadata; deliberately excluded from the partition.

Metadata example supplied in ppt_updates.md; scene is not a third partition axis.

Two attributes. Two axes.

Only two BDD100K metadata attributes define the partition.

CLIENT

weather

- clear
- overcast
- rainy
- snowy
- partly cloudy

5 clients · one per weather condition

TASK

timeofday

- 1 daytime
- 2 night
- 3 dawn / dusk

3 sequential tasks · in this order

The client × task grid

Training images per cell · darker cells contain more data (logarithmic colour scale).

	daytime	night	dawn / dusk
clear	11,375	20,927	1,849
overcast	6,872	72	1,064
rainy	2,335	1,996	307
snowy	2,628	2,018	408
partly cloudy	3,920	44	532

LARGEST CELL

20,927

clear · night

SMALLEST CELL

44

partly cloudy · night

Columns follow the sequential task order.

The setup, at a glance

TASK

Multi-Label Detection

Object detection

DATASET

BDD100K

Weather-partitioned clients

BASE MODEL

Faster R-CNN

ResNet-50-FPN

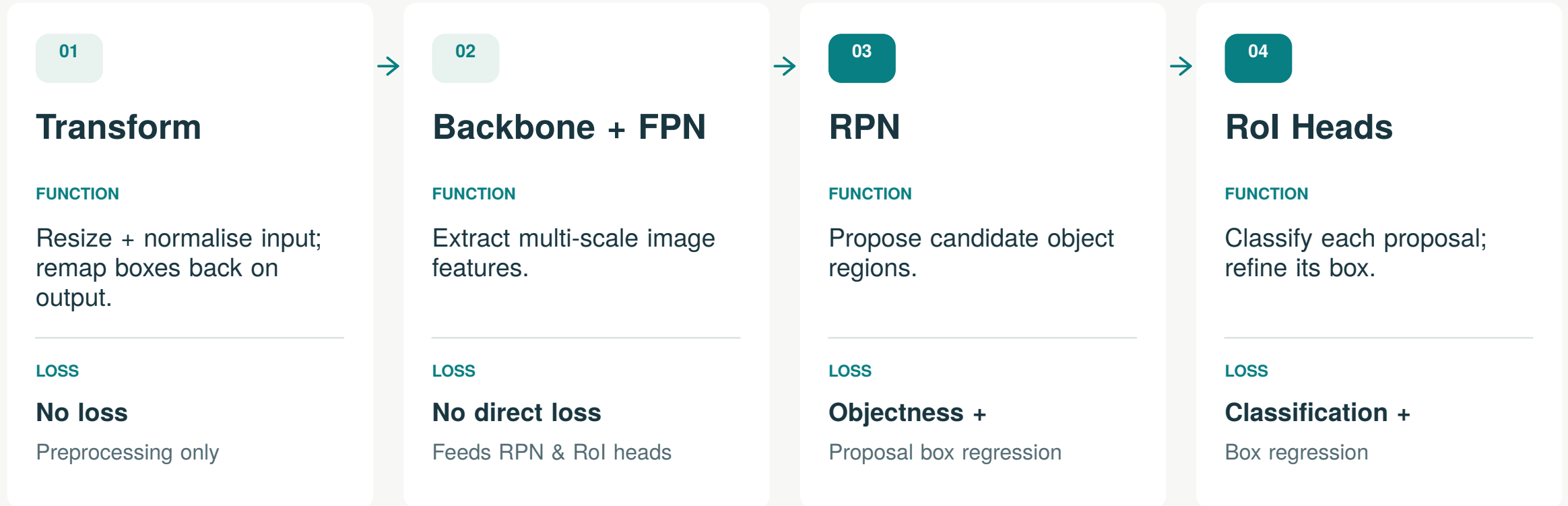
PRETRAINING

COCO

Frozen pretrained components

11 classes total = 10 BDD100K object classes + background

Model anatomy: function, then loss



Total loss = Objectness + Proposal box regression + Classification + Box regression

Source reference in original deck: [design_choices.pdf](#)

Why the backbone can stay frozen

Pretrained COCO features transfer to the overlapping BDD100K classes.

8 / 10

**BDD100K classes
already in COCO**

Shared categories
support feature transfer.

COCO $\cap$ **BDD100K**

car

truck

bus

train

person

bike

motor

traffic light

NO DIRECT COCO ANALOGUE

**traffic sign
rider**

+1 background

Added explicitly as the model's
11th class.

A long tail at the image level

Share of images containing at least one box of each class, sorted by image-level presence.

CLASS	IMAGE PRESENCE (0–100%)	%	IMAGES	INSTANCES
car		98.93	69,669	722,924
traffic sign		82.24	57,918	242,795
traffic light		56.23	39,598	188,193
person		29.06	20,469	78,121
truck		26.45	18,627	29,433
bus		12.23	8,614	11,135
bike		5.33	3,757	6,186
rider		4.63	3,264	4,103
motor		3.09	2,175	2,863
train		0.16	113	145

NEAR-UNIVERSAL / CAR

98.93%

RAREST / TRAIN

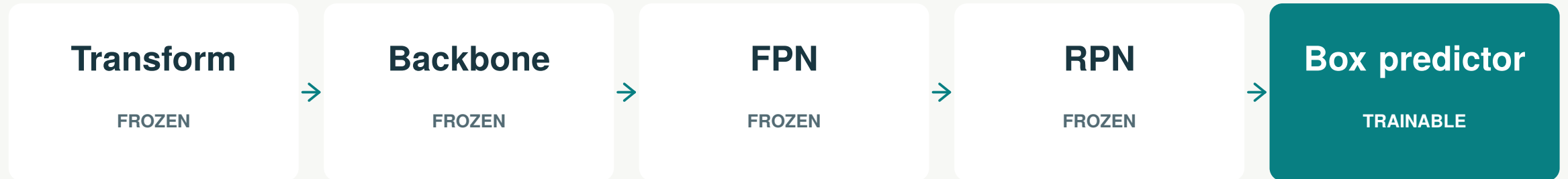
0.16%

Only 113 images. Rarity-weighted replay protects examples like these.

Pool: 70,426 usable train+val images; 5 weather clients × 3 tasks; foggy/undefined excluded.
Each image counts once per class, regardless of box count. Classes can co-occur. Statistics supplied in ppt_updates.md.

Frozen base. Trainable head.

Only roi_heads.box_predictor (class scores and box regression) is updated; all other detector weights are frozen.



LR / PRETRAINED COMPONENTS

0.0

Unused while frozen. Kept so unfreezing is a one-line edit.

LR / NEW HEAD

0.01

Learning from scratch; the active trainable component.

Optimisation hyperparameters

Epochs per task

3

Batch size

32

Optimiser

SGD

Momentum

0.9

Weight decay

5e-4

LR warm-up

None

Augmentation

None

Mixed precision (AMP)

Enabled

Replay buffer: protecting rare classes

CAPACITY

2,000

images · filenames only

REPLAY RATIO

1 : 1

current-task : replay
per batch

EVICTON SCHEME

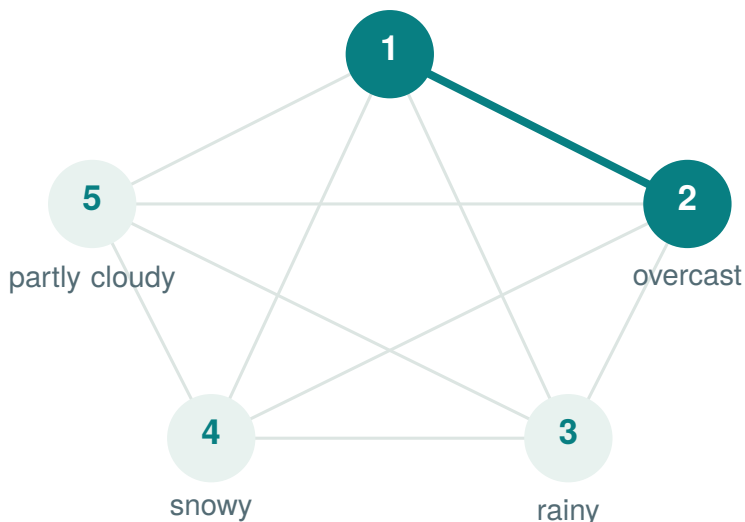
- 1 **Score each image by its rarest class**
Rarer class present → higher priority.
- 2 **When a full buffer gets a new image**
Compare its priority with the weakest entry.
- 3 **Keep the rarer, drop the more common**
Common-class images are evicted first; rare train images retain priority.

Pairwise gossip, paced by each client

Any client can contact any other; one uniformly random partner is chosen from the other four.

Fanout = 1

Fixed by design, not a FANOUT setting.
clear



Highlighted: head-only exchange. Including the frozen base would make the payload $\sim 40\times$ larger.

GOSSIP_EVERY_N_STEPS

20

local steps/client

GATE_BATCH_SIZE

32

own-data images

GOSSIP_LAMBDA

0.5

mixing ceiling

GATE_SENSITIVITY

2.0

trust decay

Gate data: own replay buffer, or the current task batch if empty. Each side evaluates on its own data.

No global rounds: more local steps mean more initiations over a task. Frozen backbone/FPN/RPN stay local.

Long intervals risk isolated task completion; short intervals add gate-evaluation cost (two models per local gate).

Implementation references supplied in ppt_updates.md: sample_partner() in gossip.py; gate_batch() in train.py; config.py.

Gate locally. Mix only the trainable head.

Compare my model and my neighbour on the same local batch, using the training loss for both.

ℓ_{own} : my loss $\ell_{\text{neighbour}}$: neighbour's loss

$$e = \max\left(0, \frac{\ell_{\text{neighbour}} - \ell_{\text{own}}}{\ell_{\text{own}}}\right)$$

Relative excess keeps gating comparable as training losses change.

$$g = \exp(-s e), \quad s = 2.0$$

$$\alpha = \lambda g, \quad \lambda = 0.5$$

e : excess g : gate_weight α : effective mixing weight

Normalised excess is defined here for $\ell_{\text{own}} > 0$.

WHAT THE TWO KNOBS CONTROL

No worse than me: $e = 0, g = 1$. Full gate strength, not full replacement.

Higher sensitivity s makes trust decay faster as normalised excess grows.

λ is the ceiling: $0 < \alpha \leq 0.5$. A worse neighbour gets a smaller share.

$$\theta_{\text{head}} \leftarrow (1 - \alpha) \theta_{\text{head}} + \alpha \theta_{\text{neighbour,head}}$$

Trainable head parameters only

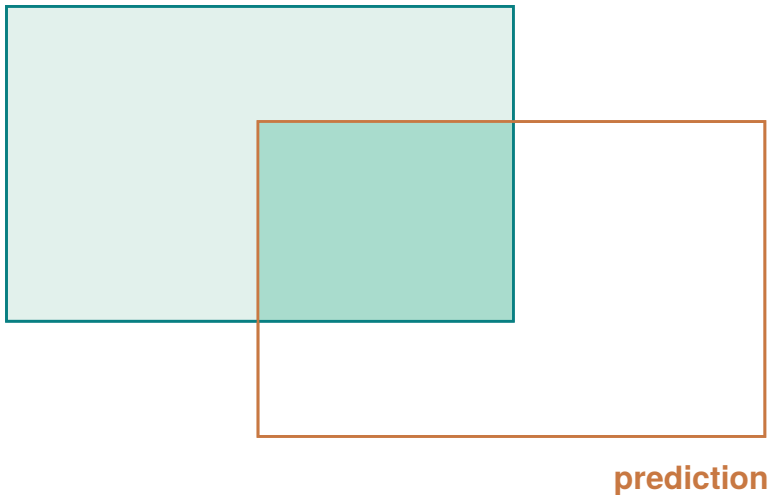
Supplied implementation references: `gate_weight()` and `mix_into()` in `gossip.py`. Each side computes its own gate.

IoU, then a verdict

A detection must clear the IoU threshold to match a ground-truth box.

INTERSECTION OVER UNION

ground truth



IoU = Overlap area / Union area

TP

Detection matched to an unclaimed ground truth above the IoU threshold

FP

Detection with no qualifying match

FN

Ground-truth box no detection claimed

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

Build the Precision–Recall curve

For one class at one fixed IoU threshold, lower the confidence cutoff by accepting predictions in score order.

1 Fix the matching rule

Choose one IoU threshold; keep the same one-to-one TP/FP matching rule.

2 Sort by confidence, descending

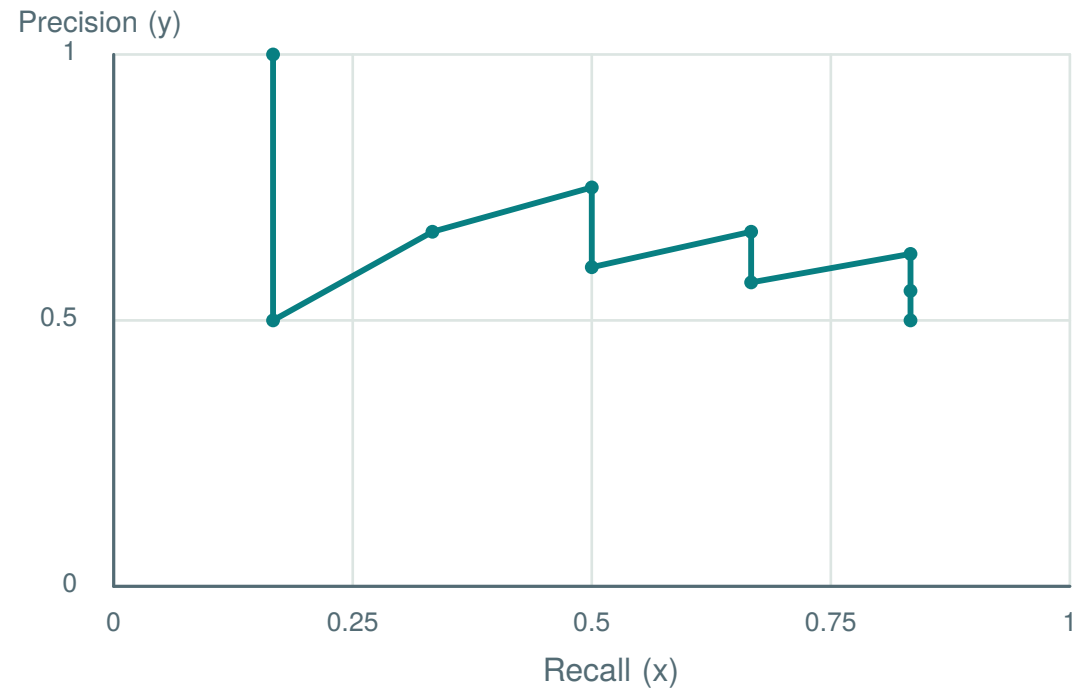
Pool this class's predictions across the evaluation set; accept them one at a time.

3 Recompute cumulative P and R

Each prefix of the ranked list gives one point on the PR curve.

Recall (x): fixed ground-truth total; TP never decreases. Precision (y) can rise or fall.

ILLUSTRATIVE RANKED PREDICTIONS / NOT RESULTS

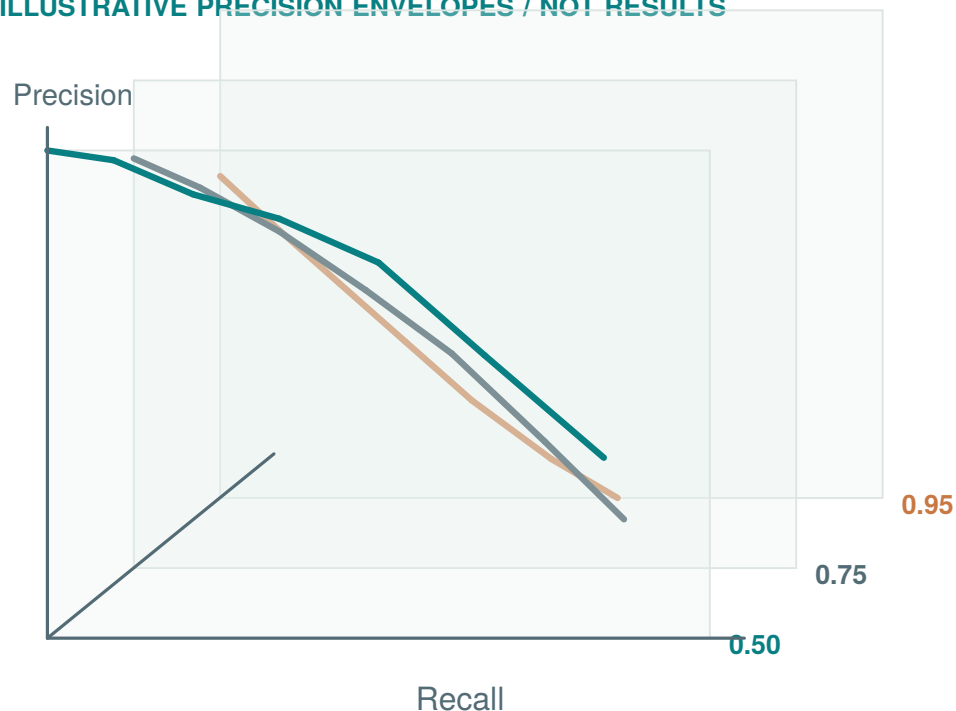


Recall need not reach 1; precision need not start high. Lower-confidence predictions often reduce precision.

IoU adds a third axis

Each fixed IoU threshold gives a separate PR curve; stricter matching tests tighter localisation.

ILLUSTRATIVE PRECISION ENVELOPES / NOT RESULTS



Depth = IoU threshold: front 0.50, middle 0.75, back 0.95

ONE SLICE: ONE CLASS, ONE IoU

AP@0.50 summarises the IoU=0.50 slice using its interpolated precision envelope.

$$AP_{\tau} \approx \int_0^1 p_{\text{interp}}(r; \tau) dr$$

TEN IoU SLICES: 0.50 TO 0.95

Average AP over 0.50, 0.55, ..., 0.95; average across object classes for the headline mAP score.

Stricter IoU changes TP/FP labels for the same predictions; precision and recall at a given cutoff tend to drop.

Three definitions. One ladder.

AP

Area under the interpolated PR curve

Per object class, per cell, at one fixed IoU



mAP

Mean of AP across the 10 object classes

Per cell



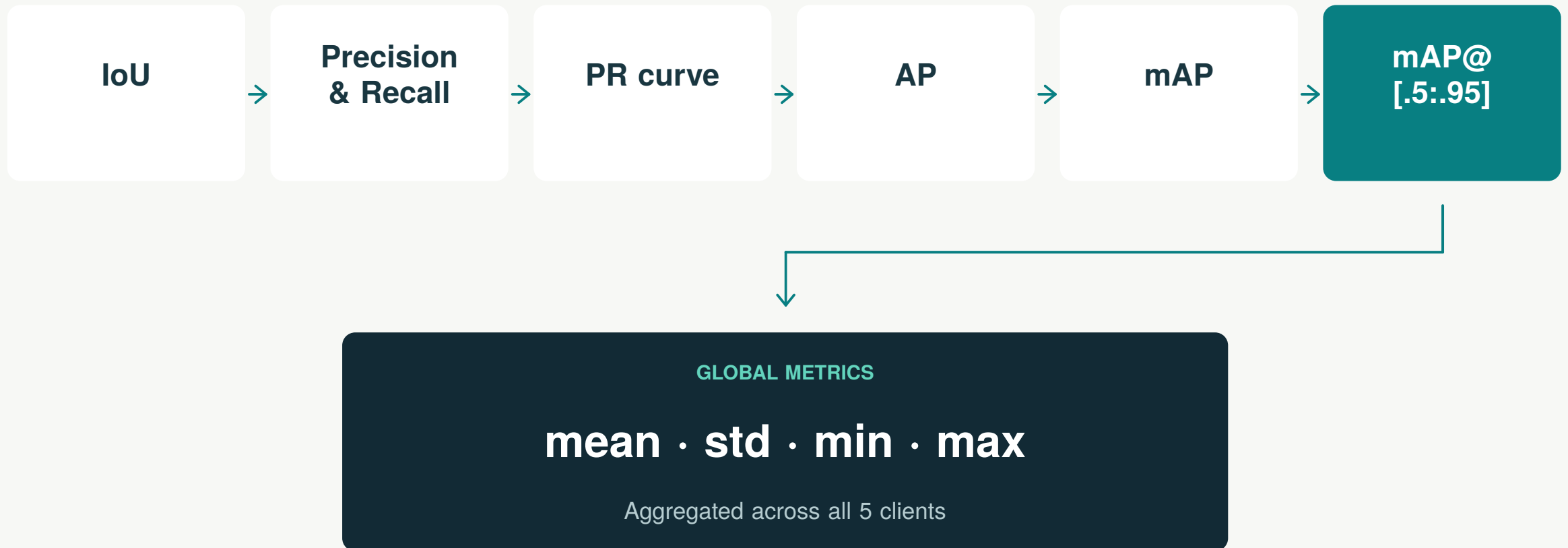
mAP@[.5:.95]

Mean of mAP across IoU thresholds 0.5 → 0.95

10 thresholds: 0.50, 0.55, ..., 0.95

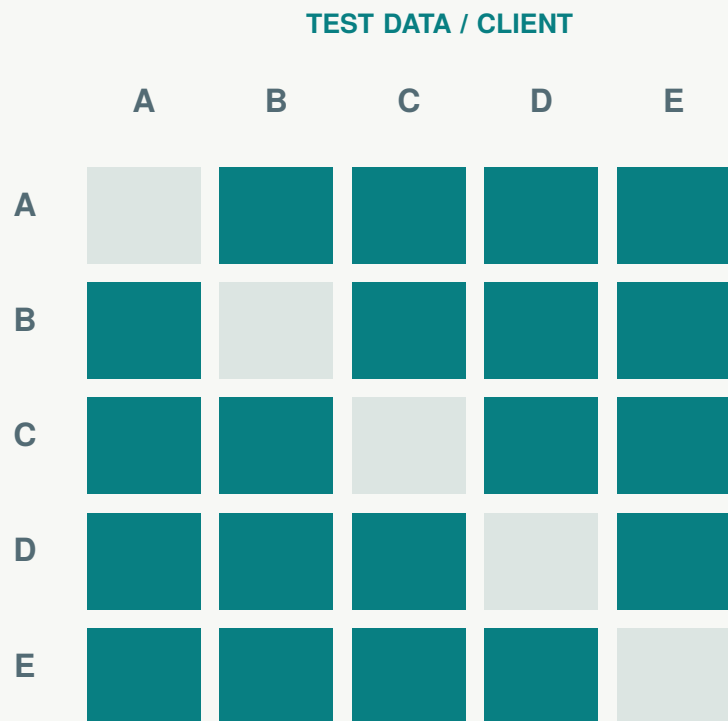
Source reference in original deck: design_choices.pdf

From one box to a global number



Did knowledge spread between clients?

Standard detection metrics alone do not capture cross-client transfer.



Rows = models · Columns = client data

HEADLINE METRIC

Global Cross-Client mAP

Off-diagonal mean: every model scored on every other client's weather condition.

Own-vs-Other Gap

Diagonal mean – off-diagonal mean.
Signed gaps can cancel; also report the mean absolute gap.
A smaller signed mean alone is not evidence of transfer.

Four runs. One controlled gossip comparison.

56,347 train / 14,079 test images; only the final RoI box predictor is trainable.

RUN	FL / GOSSIP	REPLAY	CONTINUAL	STEPS	ROLE
 gossip	Yes	Yes	Yes	8,043	Full method
 local_only	No	Yes	Yes	8,043	Replay-only control
 sequential	No	No	Yes	5,304	Naive continual baseline
 joint	No	n/a	No	5,283	Pooled, shuffled training

CLEAN A/B: GOSSIP VS LOCAL-ONLY

Same seed and schedule, 8,043 steps; only gossip differs.

WHY JOINT REPLAY IS n/a

All 15 cells pooled and shuffled;
no past task to replay.

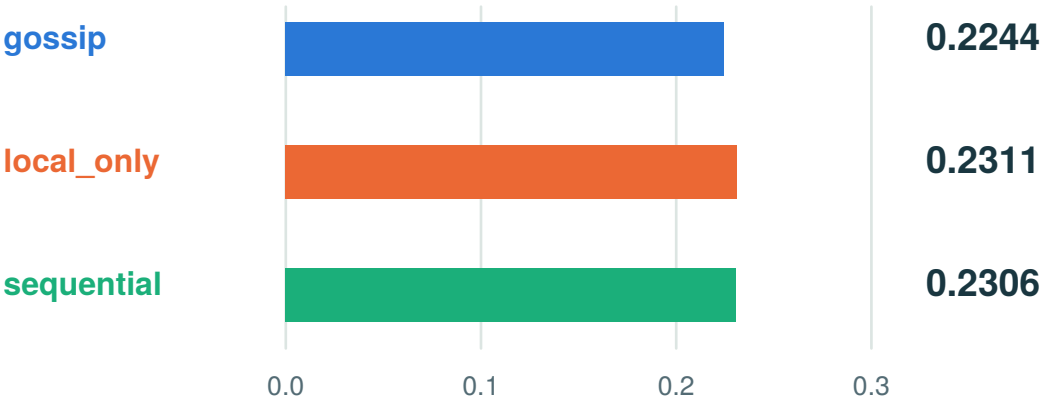
Source: analysis/summary_tables.md. FL here is pairwise gossip: no server or FedAvg. Joint steps are for one model.

Gossip trails the replay-only control

mAP@[.5:.95], averaged over the five clients. Compare runs within a panel, not across panels.

M1 / TASK-AVERAGED

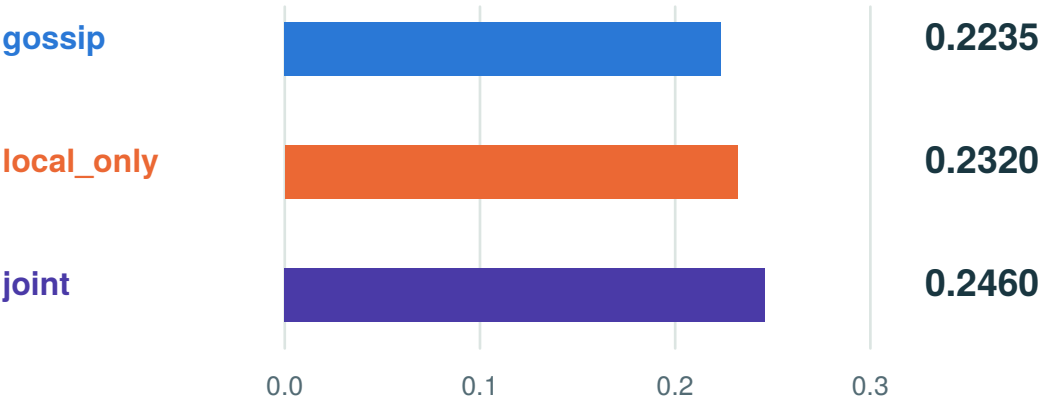
Mean of the three per-task cell scores after the final task.



Joint has no reported M1.

M2 / POOLED OWN-WEATHER

Recompute AP on each client's three test cells pooled into one evaluation set.



Sequential has no reported M2.

Gossip minus local-only: M1 -0.0067 ; M2 approximately -0.0084 (-3.6%).

Source: prism-uploads/analysis/summary_tables.md. Differences use the reported values before headline rounding.

Small measured forgetting is not a replay win

Sequential has no replay, yet its final forgetting and backward-transfer scores are also small.

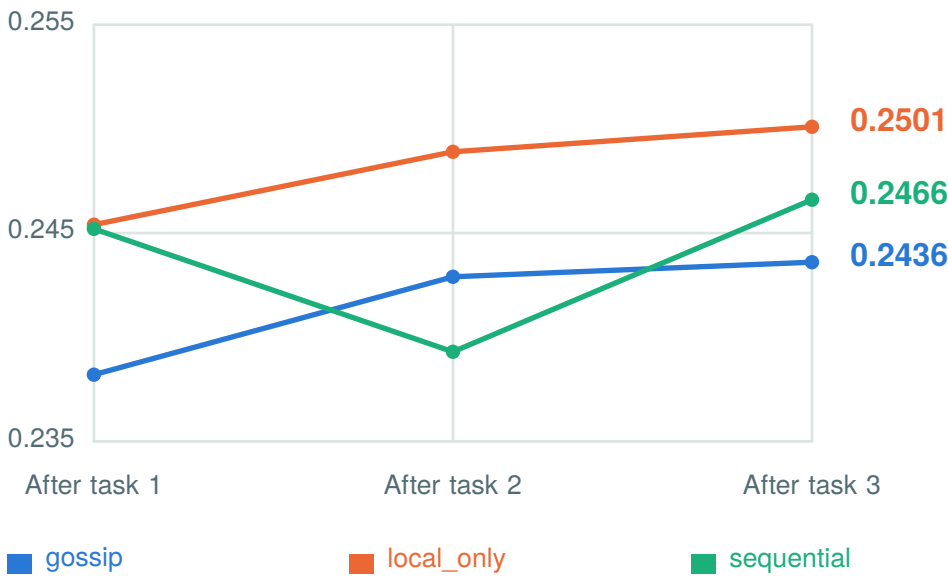
FINAL CL METRICS / MEAN OVER CLIENTS

RUN	FORGETTING	BWT
gossip	0.00061	+0.00279
local_only	0.00080	+0.00215
sequential	0.00064	+0.00251

Small averages do not establish that replay prevented forgetting or that gossip caused backward transfer.

DAYTIME RETENTION / MEAN OVER CLIENTS

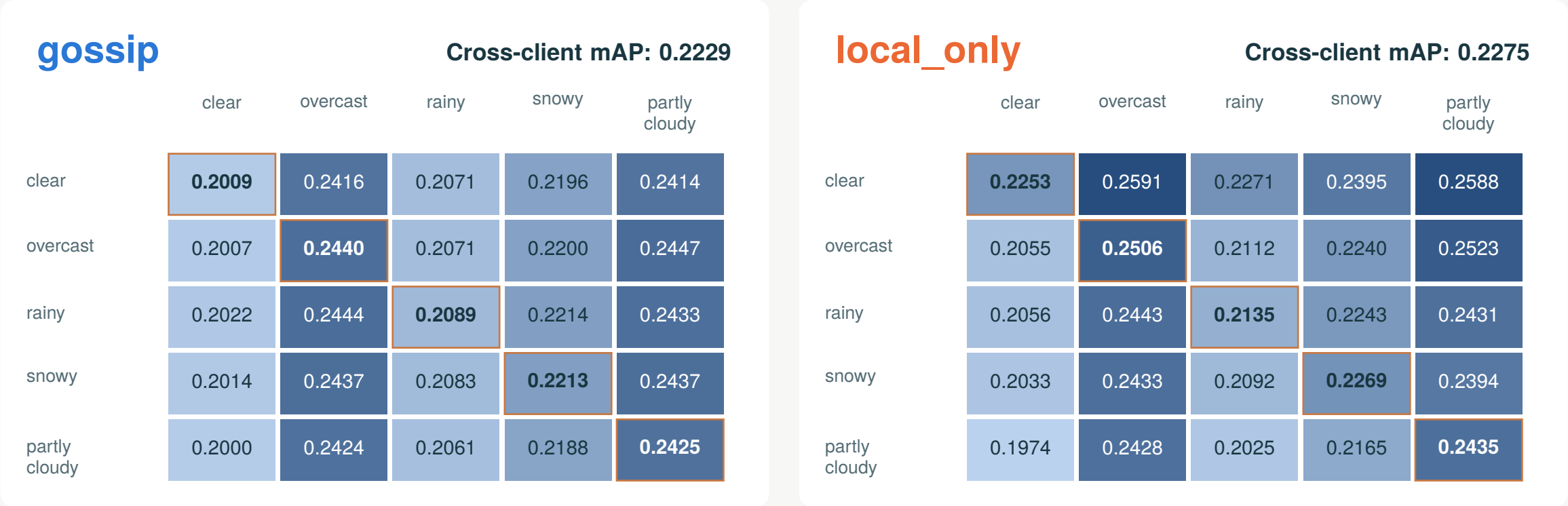
Task 1 re-evaluated after each later task.



Sources: analysis/summary_tables.md and RESULTS_BRIEF.md. Single-seed descriptive results; sequential is not step-matched.

Cross-client accuracy also decreases

Rows are client models; columns are pooled weather test sets. Both matrices use the same colour scale.



Diagonal outlined; headline = off-diagonal mean.

0.19

0.27

Local-only clear wins on all five test sets.

Source: prism-uploads/analysis/summary_tables.md. Cross-client matrices were recorded only for gossip and local_only.

More uniform scores, not better transfer

Separate agreement between models from differences in test-set difficulty.

MODEL-TO-MODEL SPREAD

Mean standard deviation within each matrix column.



~ 7.8x lower model-to-model variation, but lower cross-client mAP.

SIGNED GAPS CAN CANCEL

OWN MINUS OTHER

GOSSIP

LOCAL-ONLY

Signed mean	+0.00063	+0.00450
Mean gap	0.0200	0.0192

Report the absolute gap alongside the signed mean. A smaller signed mean alone does not demonstrate transfer.

Model scores are similar, not identical.

Source: analysis/summary_tables.md; spread ratio reported in RESULTS_BRIEF.md. Summary statistics rounded for display.

What this batch can—and cannot—tell us

A negative result for gossip in this configuration, not a general verdict on federated continual learning.

One seed, descriptive evidence

Only seed 42 is compared here. No confidence intervals or significance tests; raw run JSONs are not in this upload.

Training budgets differ

Gossip and local-only: 8,043 steps each. Sequential: 5,304 (34% fewer). Replay effects are therefore not isolated.

Tiny night test cells

Overcast/night has 18 test images; partly cloudy/night has 10. M1 gives each task cell equal weight, so interpret cautiously.

Gate behaviour was not logged

406 gossip operations, but no gate weights logged in this batch. Separate seed-7 diagnostics cannot establish a seed-42 gate effect.

In this batch, gossip lowers mAP relative to its step-matched replay-only control.

Sources: prism-uploads/analysis/summary_tables.md and RESULTS_BRIEF.md. Mechanistic explanations require further tests.

v3 / a16rpn

Detector additions and supplied result plots

BDD100K: five weather-specific clients; seed 42.

This section reports v3 separately from the earlier experiment slides.

11 supplied plots, arranged by filename and topic.

No image-based analysis or newly extracted numerical claims.

v3 detector and optimization

Smaller anchors, trainable RPN, frozen backbone

Anchors: 16, 32, 64, 128, 256 pixels; ratios: 0.5, 1.0, 2.0.

Trainable: RoI heads and RPN. Frozen: ResNet-50 and FPN.

Learning rates: head 0.01; RPN 0.001. Linear warmup: 100 steps, start 1%.

3 epochs/task; batch 32; SGD momentum 0.9; weight decay 0.0005; AMP.

Test RPN proposals: 2,000 before NMS and 2,000 after NMS.

v3 replay and gossip

Local replay and memory-gated, pull-only mixing

Replay: 2,000 image references/client; current-to-replay ratio 1:1.

Gossip: every 20 local steps; fanout 2; participation target 3 per slice.

Mixing ceiling 0.5; gate sensitivity 2.0; gate batch 32.

Head-only gate losses; RoI heads and RPN weights are shared (~14.55M).

Finished clients remain sources; only the pulling client updates.

Reported scheduling issue: sole-active clear is selected every slice.

v3 evaluation and reported scores

Previously supplied summary; not extracted from these images

Core classes: car, traffic sign, traffic light and person.

mAP over IoU 0.50:0.05:0.95; 101 recall points; maximum 100 detections.

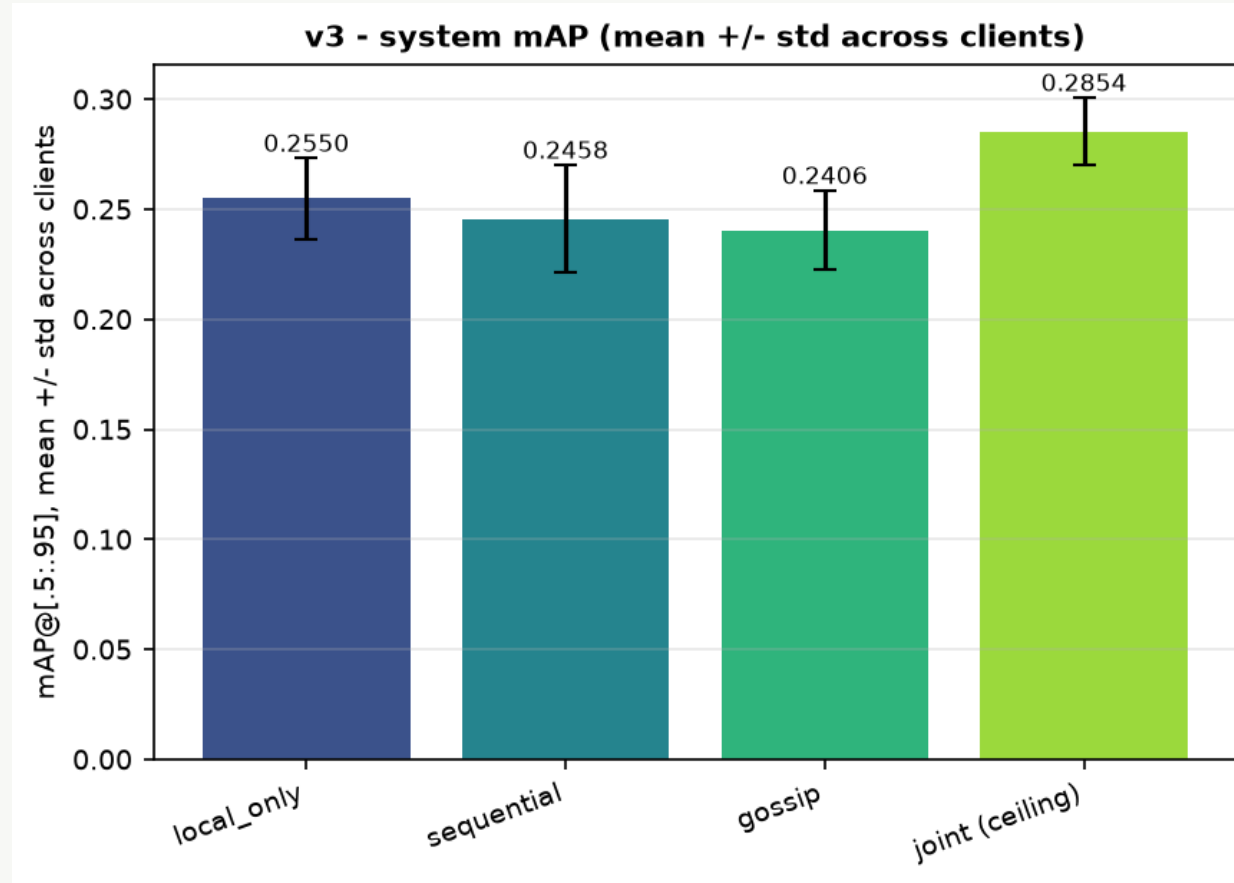
Joint 0.2854 | Local-only 0.2550 | Sequential 0.2458 | Gossip 0.2406.

P/R/F1: confidence ≥ 0.50 , IoU 0.50; class reporting floor: 50 GT.

Single seed; no uncertainty estimates or new interpretation of plots.

System-average mAP

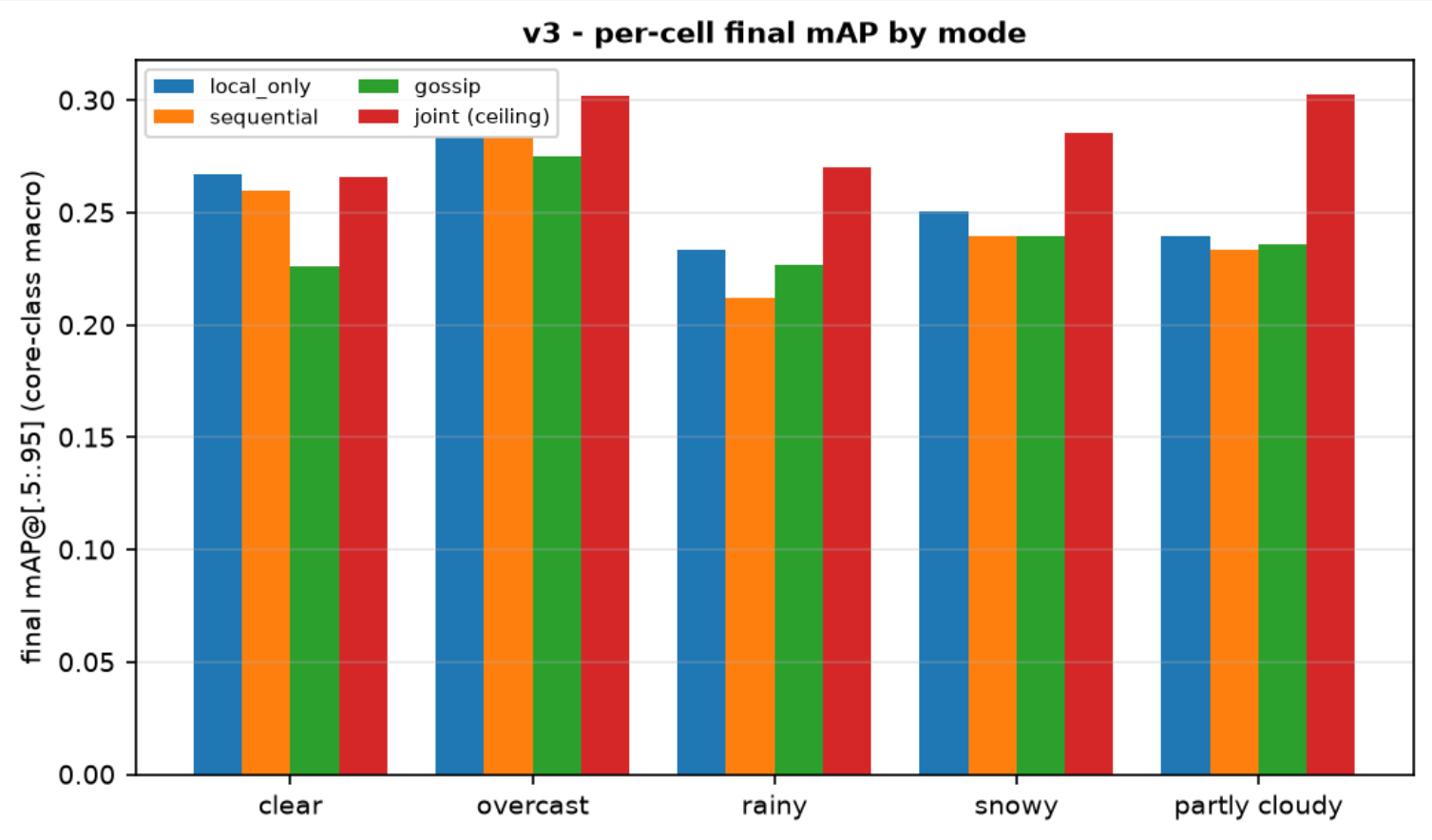
System-level results / supplied plot



Source: report_plots/a16rpn_system_avg_mAP.png

Average mAP per cell

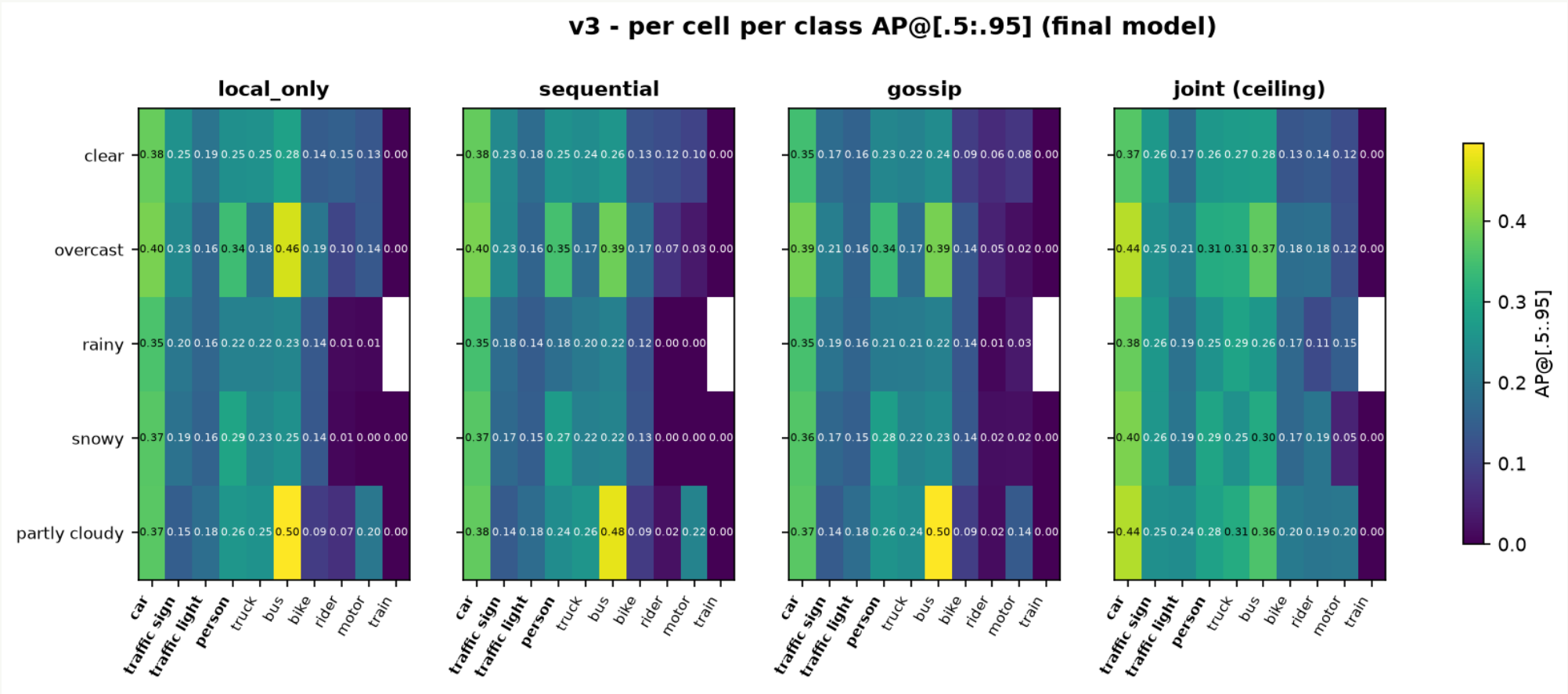
Task-cell results / supplied plot



Source: report_plots/a16rpn_per_cell_avg_mAP.png

Per-class mAP within each cell

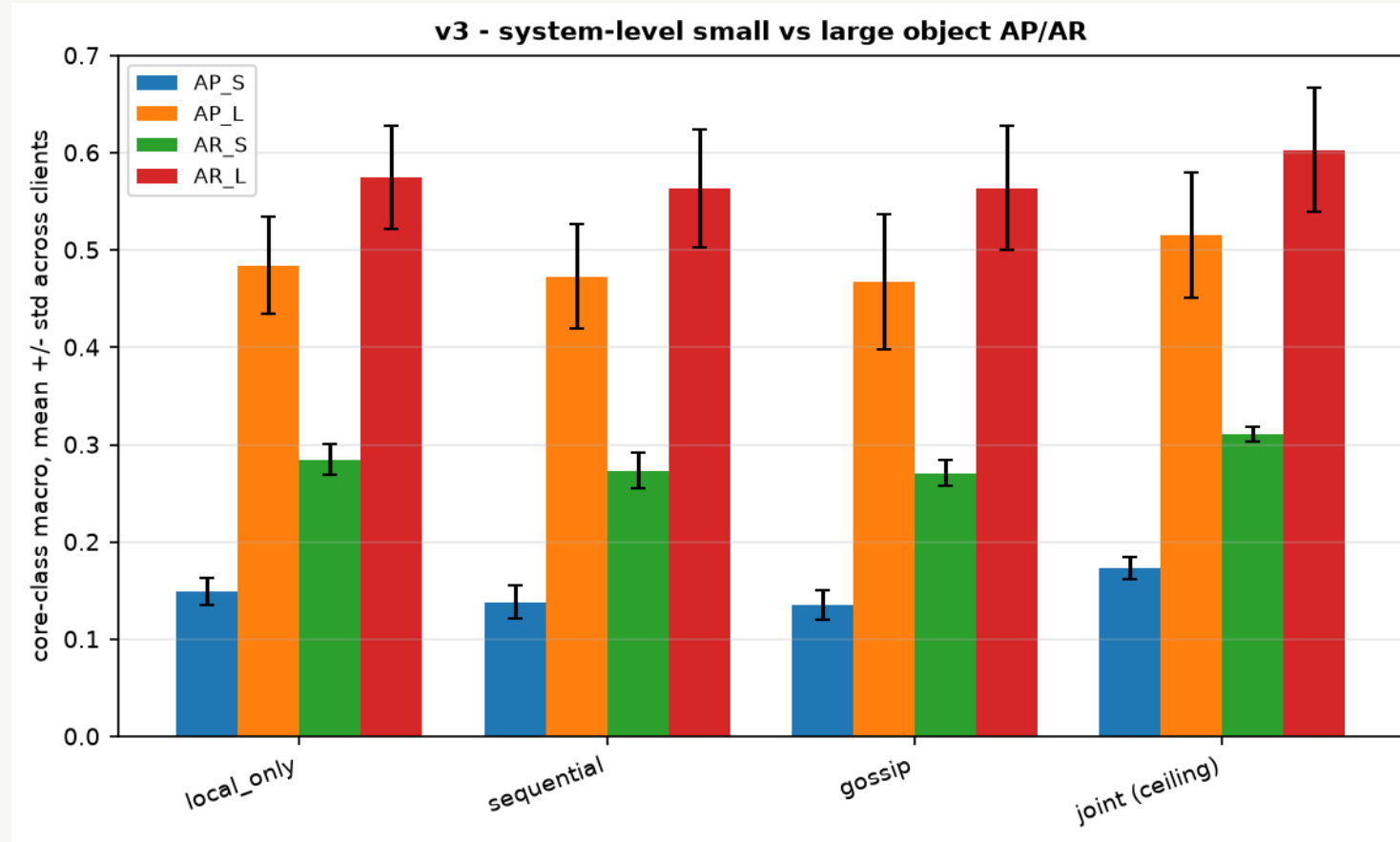
Task-cell results / supplied plot



Source: report_plots/a16rpn_per_cell_per_class_mAP.png

System-level object-size metrics

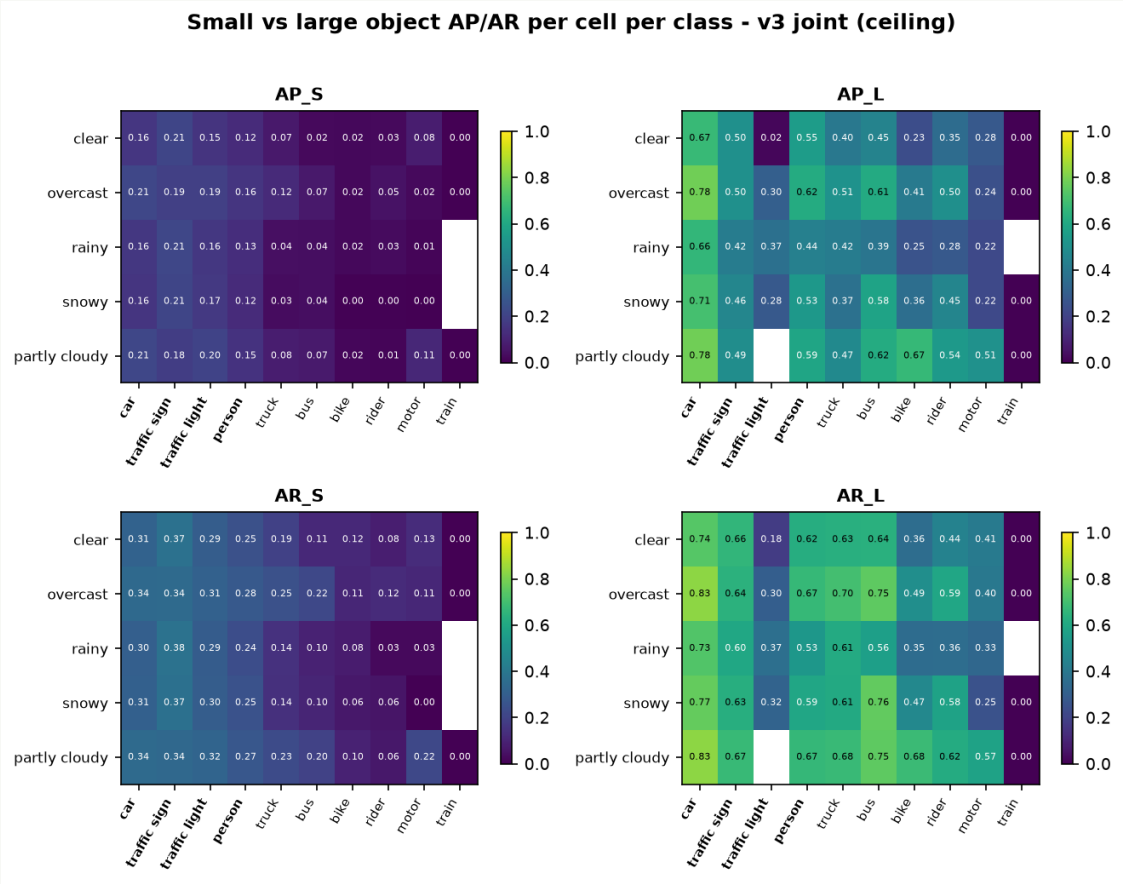
Object-size results / supplied plot



Source: report_plots/a16rpn_system_small_large_object_metrics.png

Joint: object size by class

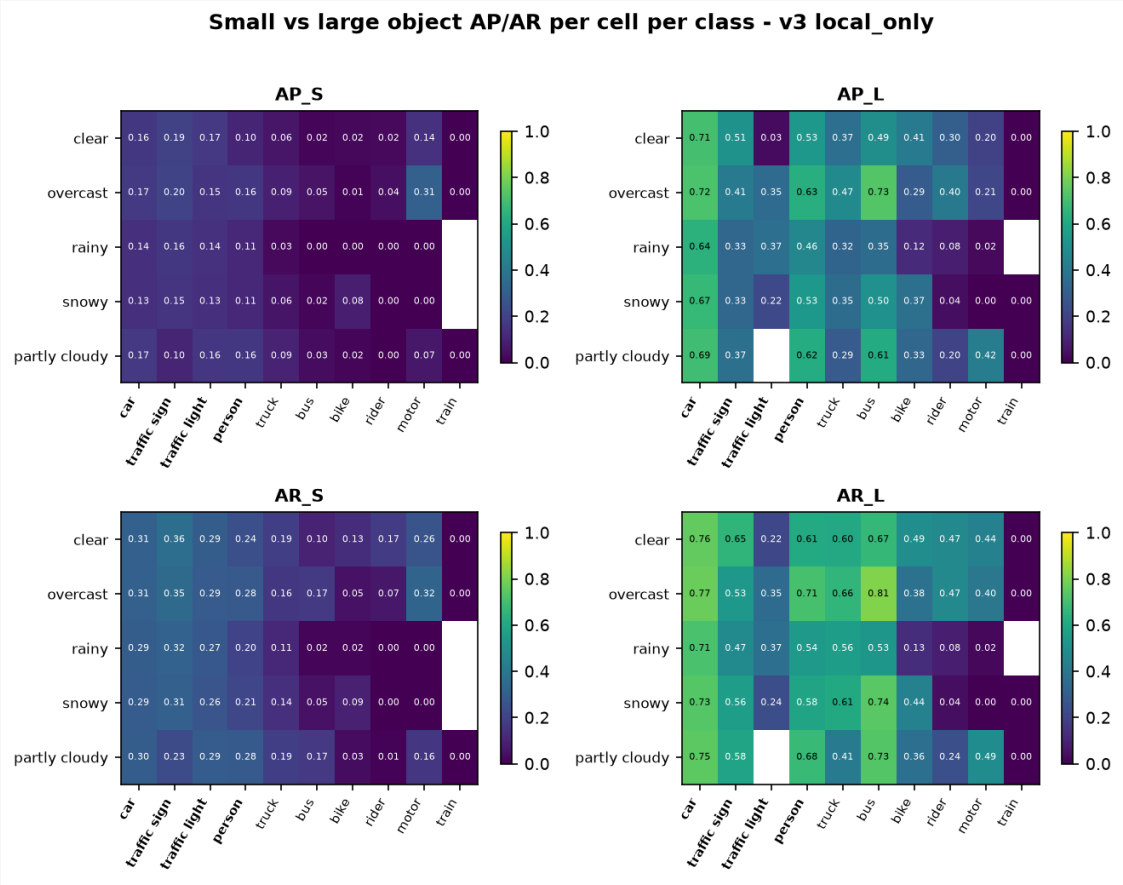
Object-size results / supplied plot



Source: report_plots/a16rpn_joint_small_large_per_class.png

Local-only: object size by class

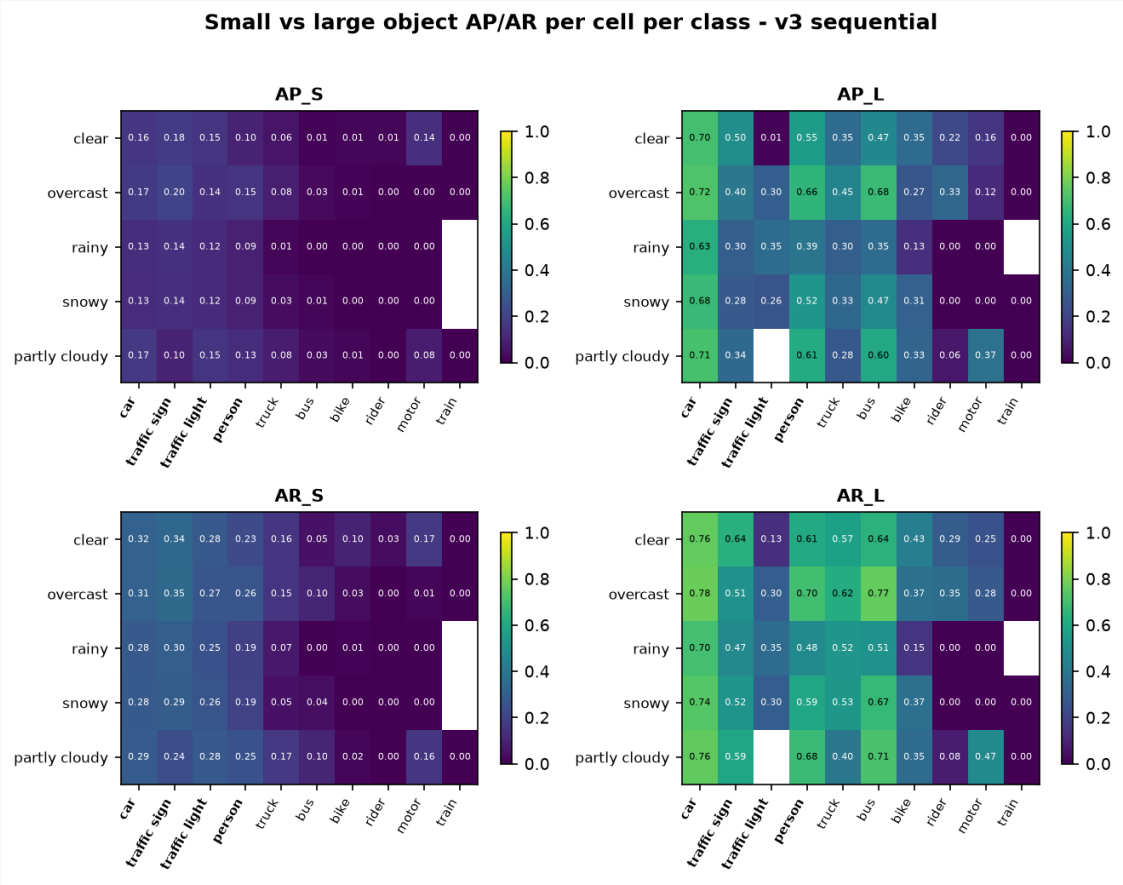
Object-size results / supplied plot



Source: report_plots/a16rpn_local_only_small_large_per_class.png

Sequential: object size by class

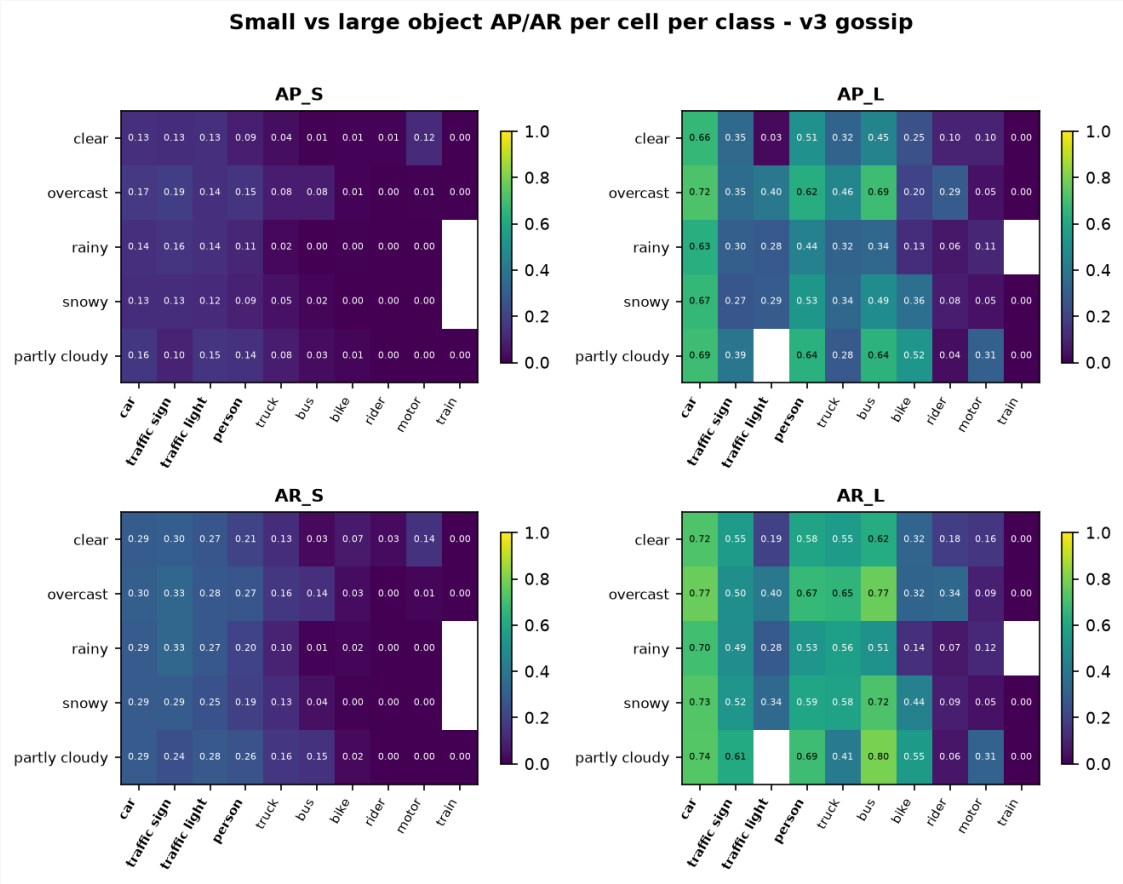
Object-size results / supplied plot



Source: report_plots/a16rpn_sequential_small_large_per_class.png

Gossip: object size by class

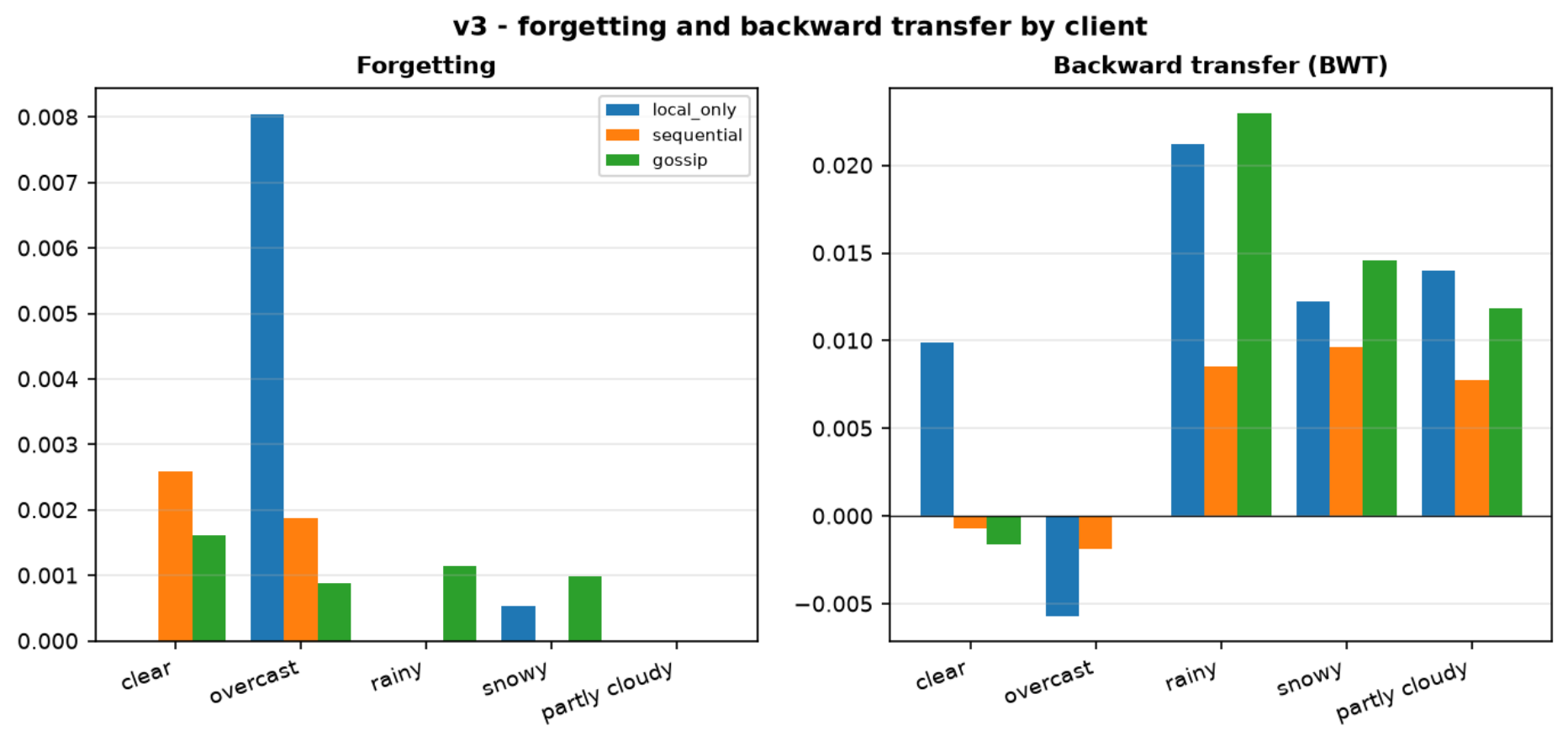
Object-size results / supplied plot



Source: report_plots/a16rpn_gossip_small_large_per_class.png

Forgetting and backward transfer

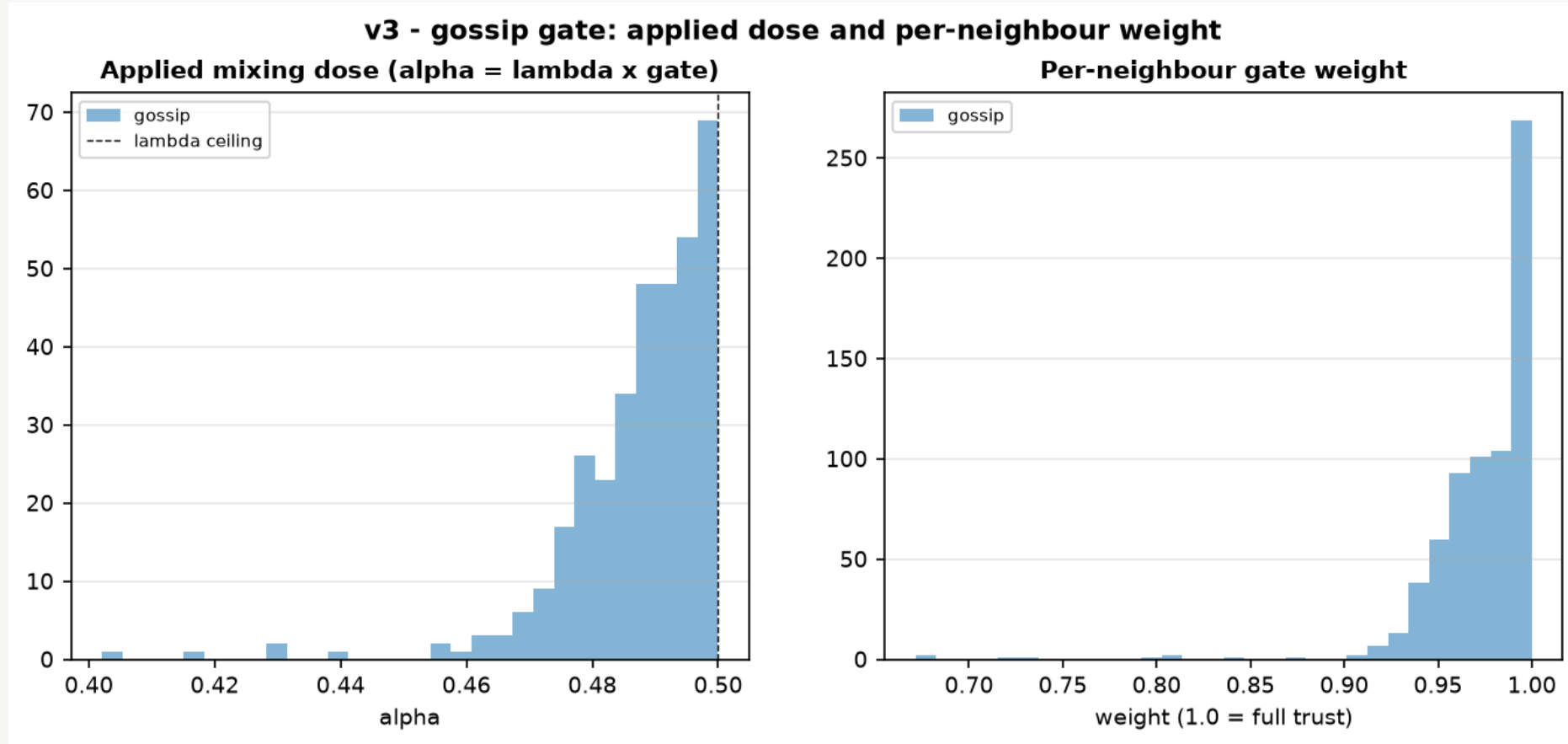
Continual-learning diagnostics / supplied plot



Source: report_plots/a16rpn_forgetting_bwt_per_client.png

Gate, alpha and weight distributions

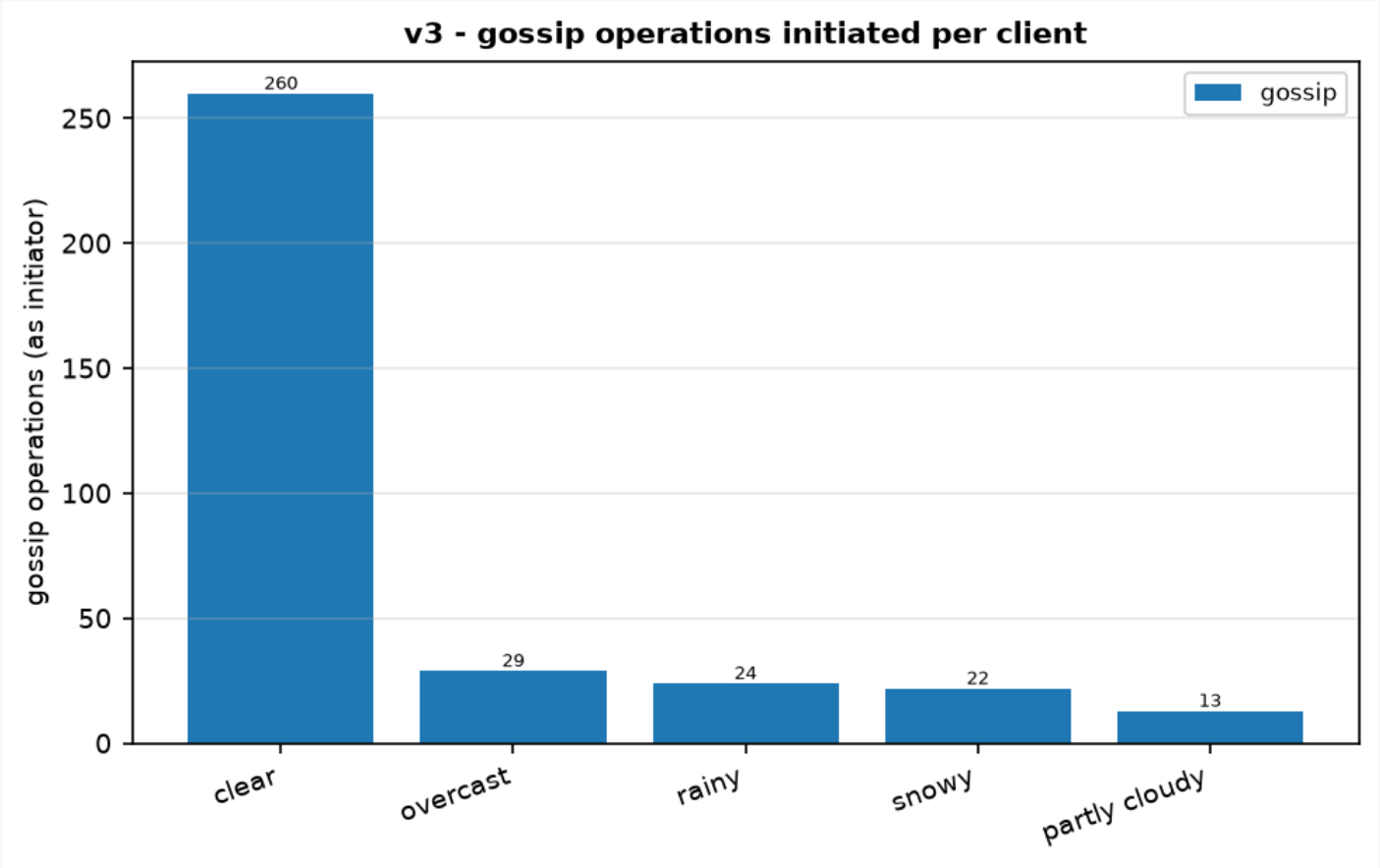
Gossip diagnostics / supplied plot



Source: report_plots/a16rpn_gate_alpha_weight_distribution.png

Gossip operations per client

Gossip diagnostics / supplied plot



Source: report_plots/a16rpn_gossip_ops_per_client.png